



KCED
Learning Augmented

24ADI003-MACHINE LEARNING

Course Objectives:

The purpose of taking this course is to:

1. Introduce the fundamental concepts of machine learning, its life cycle and ethical considerations.
2. Explore various supervised and unsupervised learning techniques and optimization strategies.
3. Examine recommendation systems, and its evaluation techniques.

Course Outcomes

- After successful completion of this course, the students shall be able to
- **CO1** Apply appropriate **data pre-processing techniques** to build machine learning models with ethical considerations. (Ap)
- **CO2** Analyze and optimize **regression models using estimation techniques, regularization, and gradient-based methods** through error analysis. (An)
- **CO3** Build and evaluate the effectiveness of **different classification models and ensemble techniques**. (E)
- **CO4** Analyze complex datasets using advanced **clustering, associative rule mining and dimensionality reduction algorithms** to uncover meaningful patterns and groupings. (An)
- **CO5** Develop **recommendation systems** to personalize user needs. (Ap)

Course Content

- **INTRODUCTION**

Introduction to Machine Learning-Types of machine learning: Supervised, Unsupervised, Semi supervised and Reinforcement Learning-Applications of machine learning in various fields- Ethics in machine learning- Fairness, accountability and interpretability -Machine learning workflow- Data Preprocessing-Feature engineering- Correlation analysis- Model training and evaluation- Model monitoring and maintenance.

- **REGRESSION MODELS**

Linear regression- Simple Regression-Least Square Estimator-Maximum Likelihood Estimator- Multiple Regression -Polynomial Regression-Performance Metrics-Bias

- **CLASSIFICATION MODELS**

Logistic Regression-Naive Bayes Classifiers-Decision Tree-K-Nearest Neighbors-Support Vector Machine – Evaluation metrics – AUC ROC- Class Imbalance –SMOTE – Cross-Validation Techniques-Ensemble Learning- Bagging- Random Forests-Boosting - AdaBoost -Gradient Boosting

- **UNSUPERVISED LEARNING**

Clustering- K-means Clustering- Gaussian Mixture Models -Hierarchical Clustering- Density-Based Clustering (DBSCAN)- Mean-Shift Clustering- Spectral Clustering- Association Rule Learning- Apriori Algorithm- FP-Growth Algorithm- Dimensionality Reduction- Principal Component Analysis (PCA)- Linear Discriminant Analysis (LDA)

- **RECOMMENDATION SYSTEMS**

Introduction to Recommendation Systems-Types- Challenges- Collaborative Filtering Techniques- User-Based- Collaborative Filtering- Item-Based- Collaborative Filtering- Matrix Factorization Techniques- Content-Based Recommendation- Hybrid Recommendation Systems- Evaluation of Recommendation Systems

Unit II – REGRESSION MODELS

Linear regression- Simple Regression-Least Square Estimator- Maximum Likelihood Estimator– Multiple Regression –Polynomial Regression-Performance Metrics-Bias Variance Tradeoff- Information Criteria-Based Model Selection- Overfitting – Underfitting – Gradient descent – Regularization – Assumptions in linear regression – Error analysis

REGRESSION

- In **Machine Learning**, regression means:
- **Learning a function that maps input variables to a continuous numerical output.**
- Formally:
- $f(X) = y$
- Where:
 - X = input features (independent variables)
 - y = output (dependent variable)
 - f = function learned from data
- Regression = predicting a number.

REGRESSION

- Everyday Examples

Problem	Input (X)	Output (y)
Predict house price	Area, bedrooms, location	₹ price
Predict salary	Experience, education	₹ salary
Predict temperature	Time, humidity	°C
Predict marks	Study hours	Marks

- All outputs are **continuous numbers** → so this is regression.

REGRESSION

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- Classification vs Regression

Task	Output Type	Example
Classification	Category	Spam / Not spam
Regression	Number	House price = ₹45,00,000

WHY REGRESSION IS A *SUPERVISED* LEARNING MODEL

- In supervised learning, the dataset contains:

(X, y) ***pairs***

- **Inputs X**
- **Correct answers y** (called **labels** or **targets**)
- The model learns by being *shown the correct answers*.
- Example(Study Hours $\rightarrow$ Marks)

Student	Study Hours (X)	Marks (y)
1	2	45
2	4	55
3	6	70
4	8	85

- Inputs = Study Hours
- Output = Marks
- Both are given $\rightarrow$ supervised learning.

CORE TERMS IN REGRESSION

- The regression model learns a function:

$$\hat{y} = f(x)$$

such that:

$$\hat{y} \approx y$$

1. Feature (Input Variable)

- A **feature** is an input used to make predictions.
- Examples:
 - Study hours
 - Attendance
 - Experience
 - Area of house
- Notation:

$$x_1, x_2, \dots, x_d$$

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CORE TERMS IN REGRESSION

2. Target / Label (Output Variable)

- The value we want to predict.
- Examples:
 - Marks
 - Salary
 - Price

Notation:

 y

3. Dataset

- A collection of input–output pairs:

$$\{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$$

CORE TERMS IN REGRESSION

4. Model

- A **model** is a mathematical function:

$$\hat{y} = f(X; \theta)$$

Where:

- θ = parameters (weights & bias)

5. Parameters (Weights & Bias)

- These are numbers learned from data.
- They control:
 - Slope
 - Shape
 - Position of the function

6. Prediction

$$\hat{y} = f(x)$$

- Model's estimated output.

CORE TERMS IN REGRESSION

7. Error (Residual)

- Difference between actual and predicted:

$$e = y - \hat{y}$$

8. Loss Function

- A function that measures how wrong the model is.
- Example (squared error):

$$L = (y - \hat{y})^2$$

9. Training

- Finding parameter values that minimize total loss.

LINEAR REGRESSION OBJECTIVE

- Learn a function that best explains the relationship between inputs and output.

Graphically:

- **Points = data**
- **Curve / line = model**
- Linear Regression is:
 - Simplest
 - Interpretable
 - Has closed-form math
 - Foundation of all regression models

The **linear regression model** is written as:

$$y = w_0 + w_1x$$

Prediction = base value + (effect × input)

LINEAR REGRESSION OBJECTIVE

- **Example:** Auto fare rule:

$$\text{Fare} = \text{Base charge} + (\text{cost per km} \times \text{distance})$$

Auto Fare	Linear Regression
Base charge	(w_0) (bias)
Cost per km	(w_1) (weight)
Distance	(x)
Fare	(y)

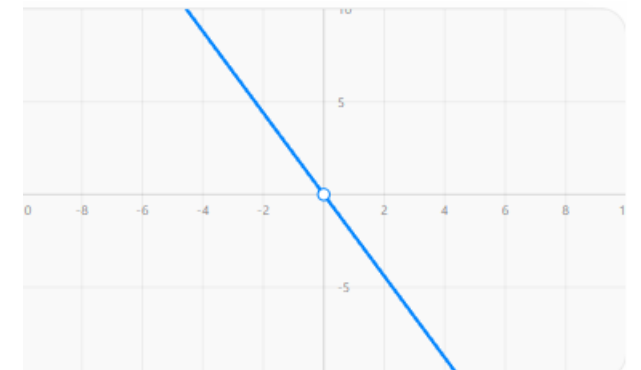
LINEAR REGRESSION OBJECTIVE

Symbol	Meaning	Description
(x)	Input / Feature	What we know (study hours, experience, area)
(y)	Output / Target	What we want to predict (marks, salary, price)
(w_1)	Weight / Slope	How strongly x affects y
(w_0)	Bias / Intercept	Starting value of y

When you draw the equation on a graph:

- x-axis $\rightarrow$ input (study hours)
- y-axis $\rightarrow$ output (marks)
- The equation draws a **straight line**

That's why it's called **linear** regression.



$$\text{Prediction} = \text{base value} + (\text{effect} \times \text{input})$$

LINEAR REGRESSION

WEIGHT w_1 (Slope)

Definition

w_1 tells us:

How much y changes when x increases by 1 unit.

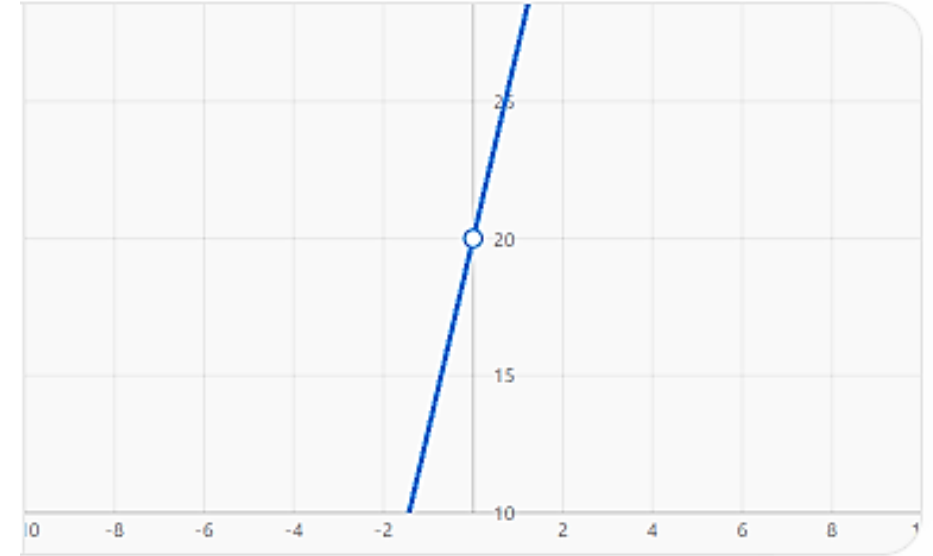
$$y = 20 + 7x$$

Then:

- Increase study hours by 1 → Marks increase by 7

Study Hours	Predicted Marks
1	27
2	34
3	41

The line goes upward → positive slope.
Sign of the Weight



(w_1) value	Meaning
$(w_1 > 0)$	As $x \uparrow$, $y \uparrow$
$(w_1 < 0)$	As $x \uparrow$, $y \downarrow$
$(w_1 = 0)$	x has no effect

LINEAR REGRESSION

BIAS w_0 (Intercept)

Definition:

w_0 is the value of y when $x = 0$.

$$y = 20 + 7x$$

If:

- $x = 0$ hours studied

Then:

$$y = 20$$

So even without studying, the model predicts **20 marks**.

Why Bias Is Needed

Imagine no bias:

$$y = 7x$$

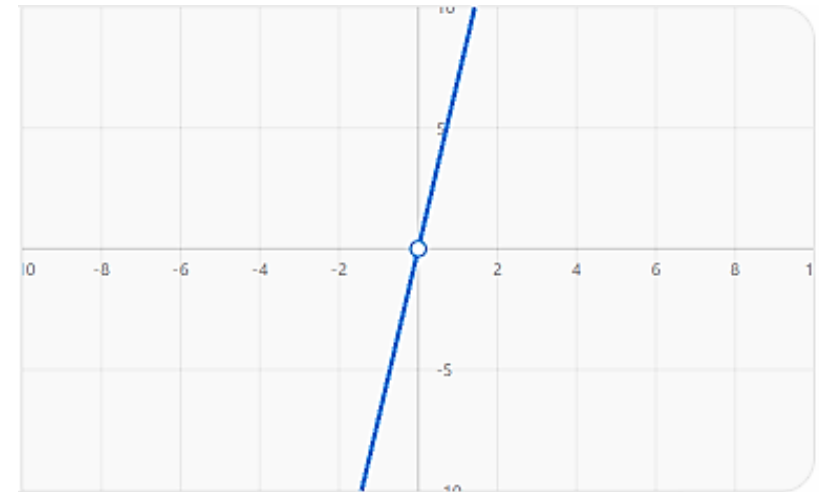
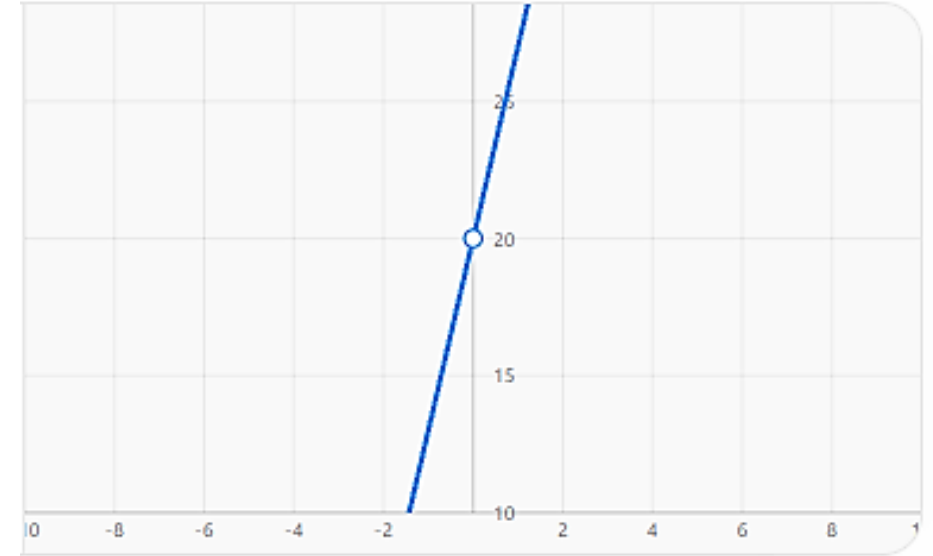
Then at $x = 0$:

- $y = 0$ (always)

But real life rarely starts from zero.

- Students may score marks without studying
- Salary exists even with 0 experience
- House has value even with 0 renovation

Bias shifts the line up or down to fit reality.



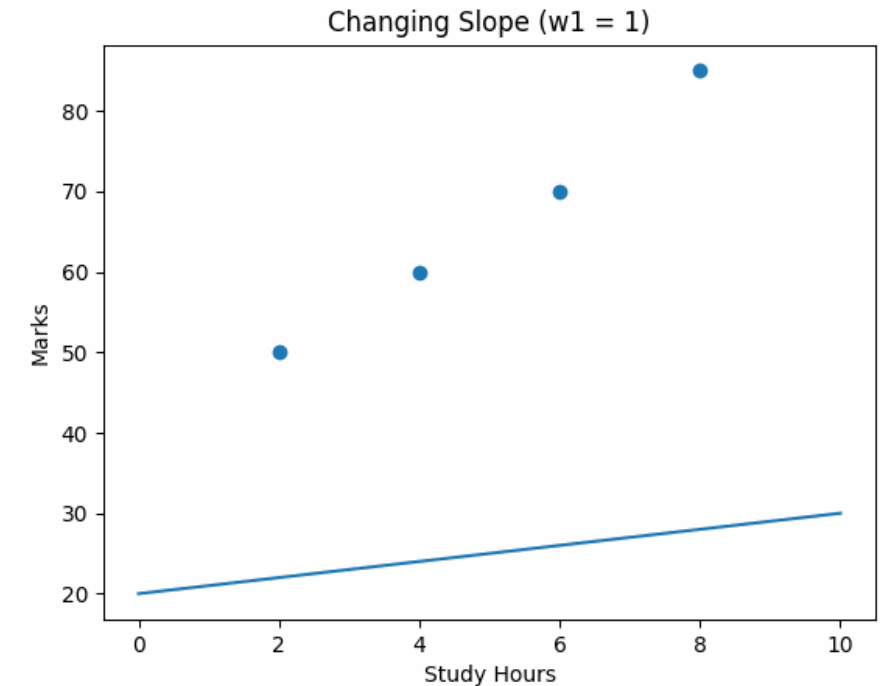
LINEAR REGRESSION

How the Line Changes When Weights Change: Case 1: Change ONLY w_1 (Slope)

Effect	What happens
Increase (w_1)	Line becomes steeper
Decrease (w_1)	Line becomes flatter

The importance of x increases or decreases.

- Increasing $w_1 \rightarrow$ line tilts more steeply
- Decreasing $w_1 \rightarrow$ line flattens
- Interpretation: *importance of input feature x changes*



LINEAR REGRESSION

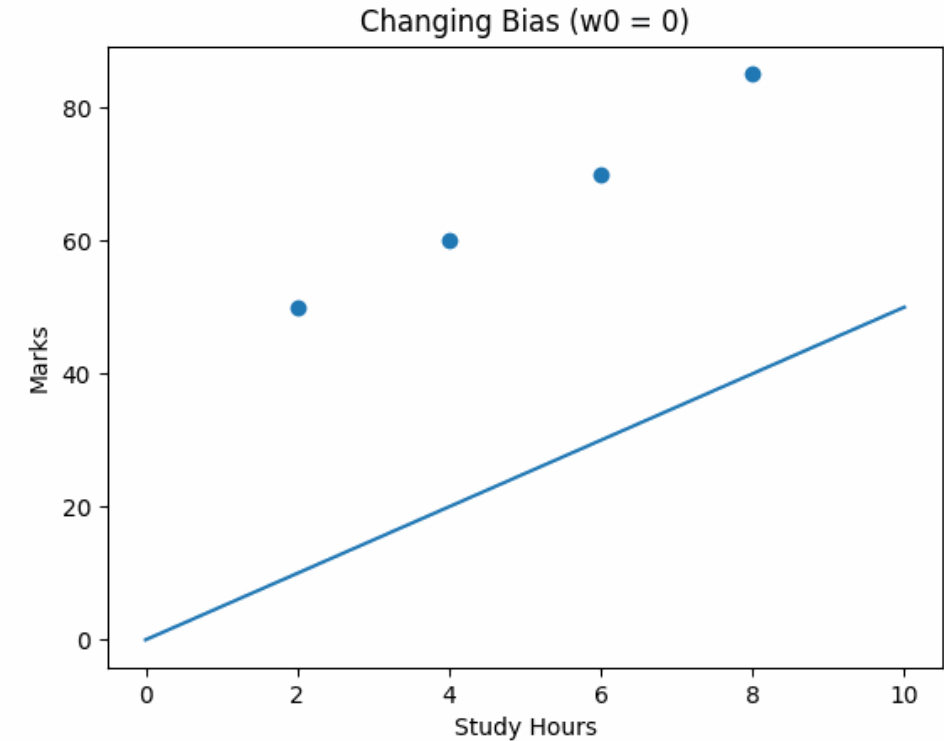
How the Line Changes When Weights Change:

Case 2: Change ONLY w_0 (Bias)

Effect	What happens
Increase (w_0)	Line moves up
Decrease (w_0)	Line moves down

base prediction changes.

- Increasing $w_0 \rightarrow$ line shifts upward
- Decreasing $w_0 \rightarrow$ line shifts downward
- Interpretation: *baseline prediction changes*



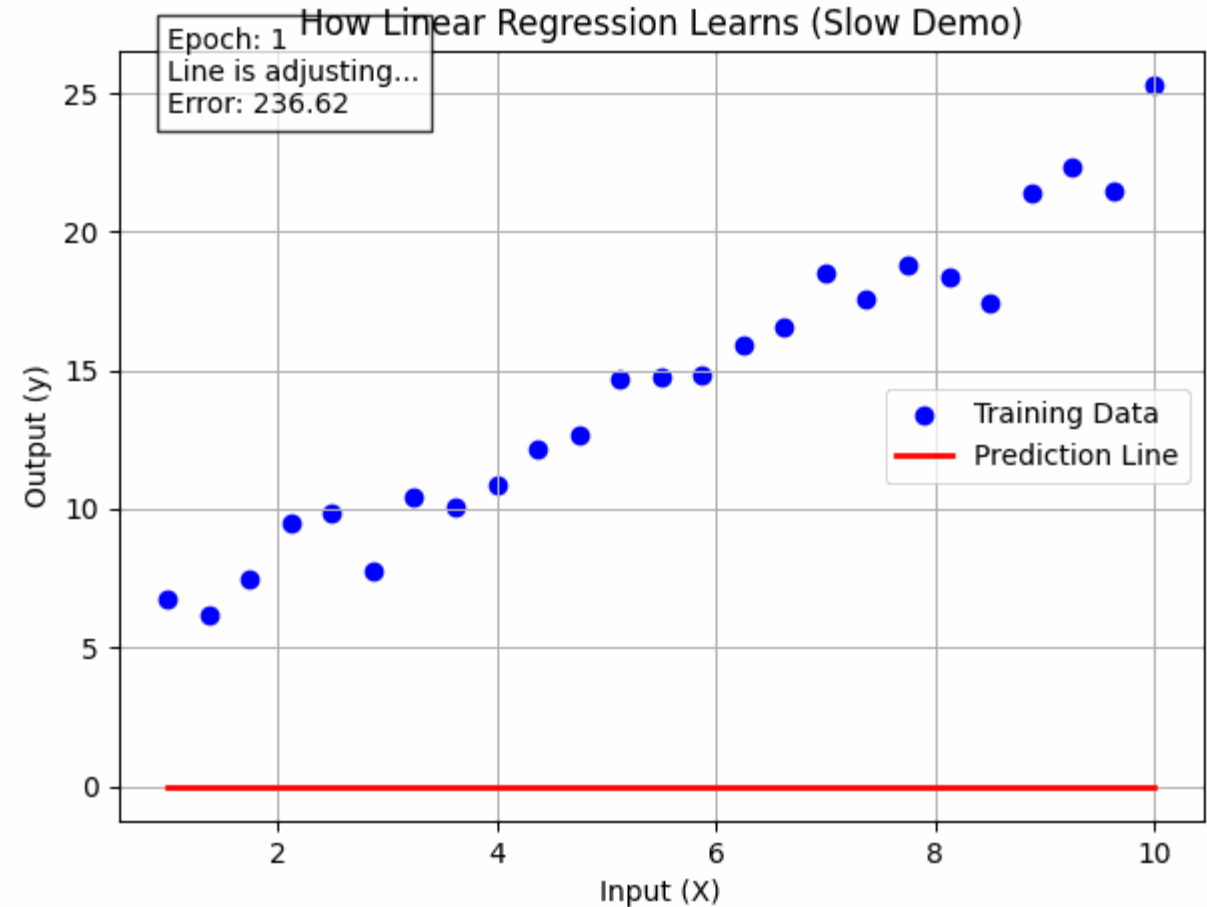
LINEAR REGRESSION

In ML training:

- We **do not fix** w_0, w_1
 - The algorithm **learns** them from data
 - Each update changes the line
- So:

Changing weights = changing the model itself

This is why your infographic shows the line moving.



LINEAR REGRESSION

Why “Best Line”? → Least Squares Idea

For each data point:

$$\text{Error} = y_i - \hat{y}_i$$

If we simply add errors, positives and negatives cancel

So we **square them**:

$$\text{Squared Error} = (y_i - \hat{y}_i)^2$$

Objective Function (Cost Function)

Least Squares Estimator finds w_0, w_1 that minimize this sum.

LINEAR REGRESSION

Example
Given Dataset (Study Hours vs Marks)

Student	x (Hours)	y (Marks)
1	1	50
2	2	55
3	3	65
4	4	70
5	5	75

Step 1: Compute Required Sums

Quantity	Calculation	Value
(n)	—	5
($\sum x$)	1+2+3+4+5	15
($\sum y$)	50+55+65+70+75	315
($\sum x^2$)	1+4+9+16+25	55
($\sum xy$)	50+110+195+280+375	1010

LINEAR REGRESSION

Example

Step 2: Calculate Slope w_1

$$w_1 = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$

$$w_1 = \frac{5(1010) - 15(315)}{5(55) - 15^2} = \frac{5050 - 4725}{275 - 225} = \frac{325}{50} = 6.5$$

Step 3: Calculate Bias w_0

$$w_0 = \frac{\sum y - w_1 \sum x}{n}$$

$$w_0 = \frac{315 - 6.5(15)}{5} = \frac{315 - 97.5}{5} = 43.5$$

Final Regression Equation

$$\hat{y} = 43.5 + 6.5x$$

LINEAR REGRESSION

Example

Step 4: Prediction

If student studies **6 hours**:

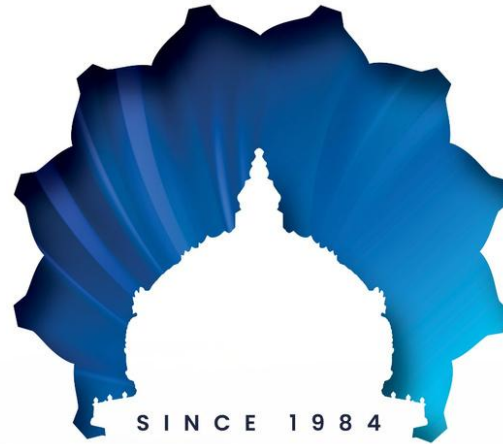
$$\hat{y} = 43.5 + 6.5(6) = 82.5$$

Slope $w_1 = 6.5$

Each additional hour of study increases marks by **~6.5**

Bias $w_0 = 43.5$

Baseline marks even with **zero study**



FOUR DECADES OF NATION BUILDING THROUGH EXCELLENCE IN EDUCATION

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